

Delays cost US airlines ~\$50B, and it's getting worse

Every US domestic flight, 2021–2025: on-time performance, delay cost and what causes it

01 The Problem

02 Where Chains Start

03 The Fix

FLIGHTS ANALYZED

33.65M

US domestic, Jan 2021 – Dec 2025

EST. AIRLINE DELAY COST

\$49.6B

5 years, in 2025 dollars

ON-TIME RATE, 2025

77.7%

Worst of the 5 years (82.8% in 2021)

DELAY FROM LATE AIRCRAFT

38.8%

Share of all delay minutes (BTS cause codes)

On-time performance by month

% of operated flights arriving less than 15 min late



Estimated delay cost by year

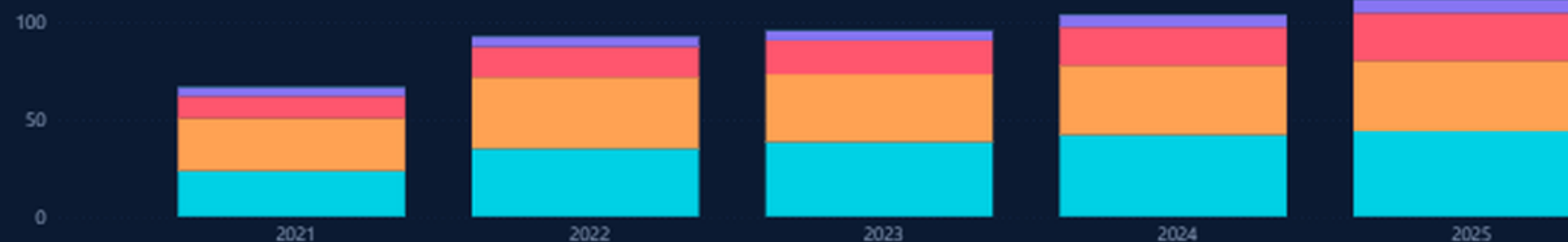
Arrival delay minutes x \$98.41 = \$ billions



What causes the delay minutes?

Delay minutes by BTS cause, millions per year

● Late aircraft ● Airline(carrier) ● Air traffic system ● Weather ● security



Key takeaways

- On-time rate fell from 82.8% (2021) to 77.7% (2025), the worst year of the period.
- 2025 cost more than 2024 despite fewer flights: delay cost per flight rose ~8%.
- 38.8% of delay minutes come from late-arriving aircraft: delays spreading plane to plane.

DFW is where America's costliest delay chains begin

Each flight linked to its aircraft's previous leg by tail number: 4.1M delay chains traced

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DELAY CHAINS TRACED

4.1M

Runs of consecutive late legs, same aircraft

DELAY MINUTES INHERITED

41.5%

Validated vs BTS late-aircraft share (38.8%)

LATE-MORNING SPREAD RATE

52%

Chance a 9 AM–12 PM delay spreads onward

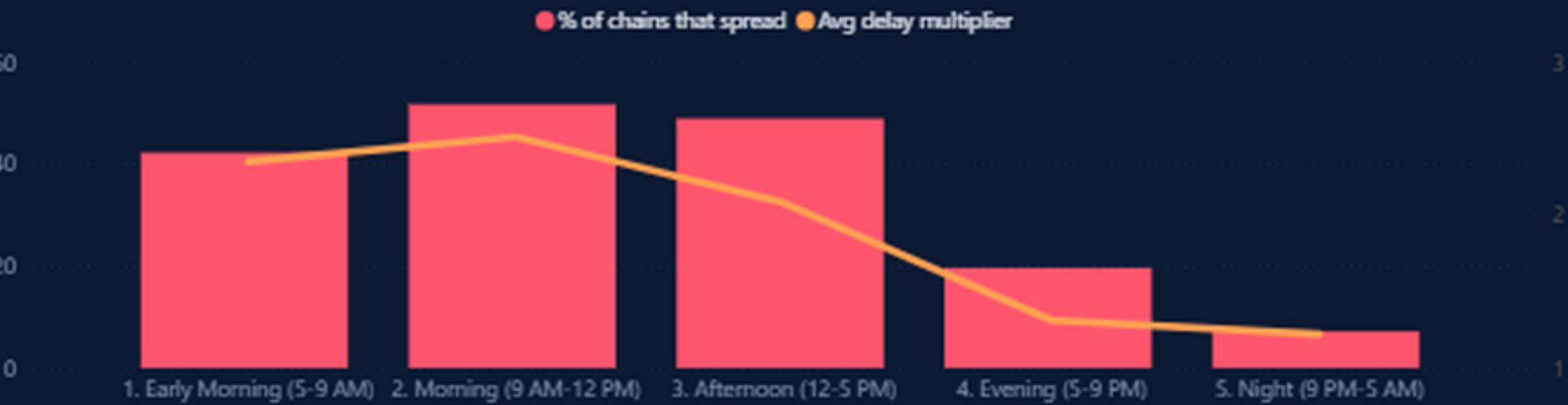
#1 ORIGIN OF CHAINS

DFW

\$2.88B in chain cost, 2021–2025

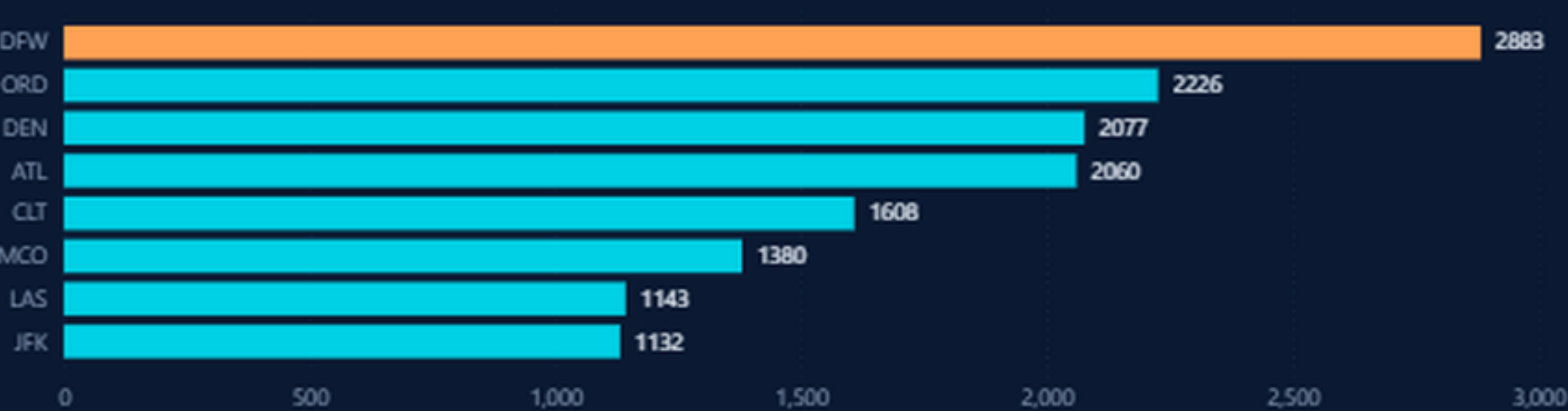
When a delay starts decides how far it spreads

% of chains that spread, by start time of the first delay



Top airports by delay-chain cost

Chain cost, \$ millions, 2021–2025



Chain cost per flight by airline

\$ of delay-chain cost per operated flight, 2021–2025



Top hotspots: airport x time of day

Downstream delay cost, \$M: what stopping the spread could save



Blanket buffers lose \$20M a year. Targeted ones nearly break even.

American Airlines turnarounds at DFW · strategy learned on 2024, tested out-of-sample on 2025

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DELAYS PASSED ON: AA @ DFW

64.0%

vs 56.3% for Delta @ ATL ($p < 0.001$)

BUFFER ABSORPTION

0.57

Delay min absorbed per buffer min (regression)

BLANKET 75-MIN RULE

-\$20.2M

Per year: pays on 74K turns, helps 13K

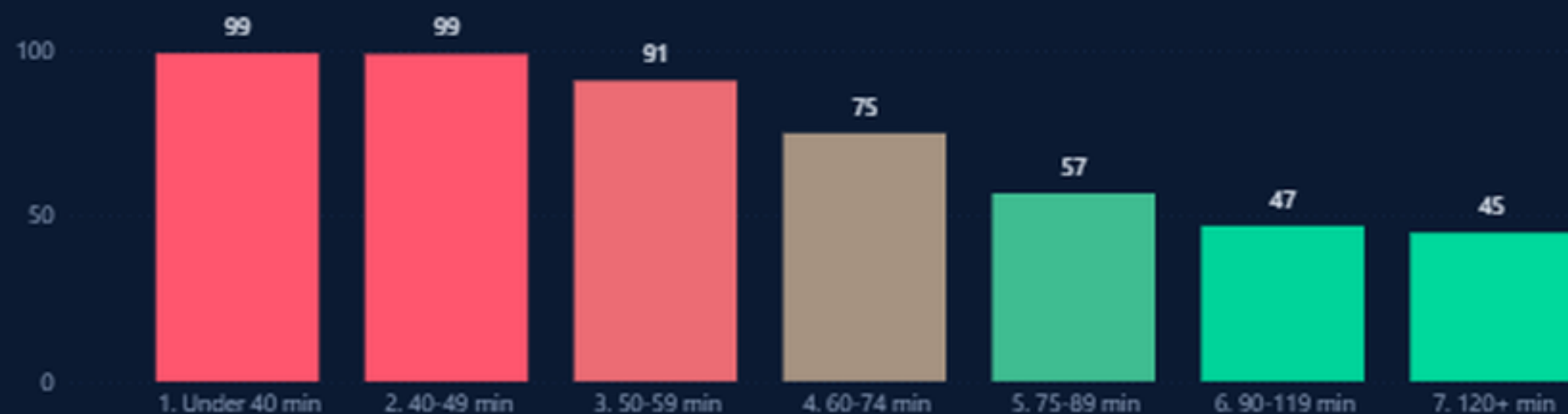
RISK-TARGETED BUFFERS

-97%

Loss cut; break-even \$17.99 per ground min

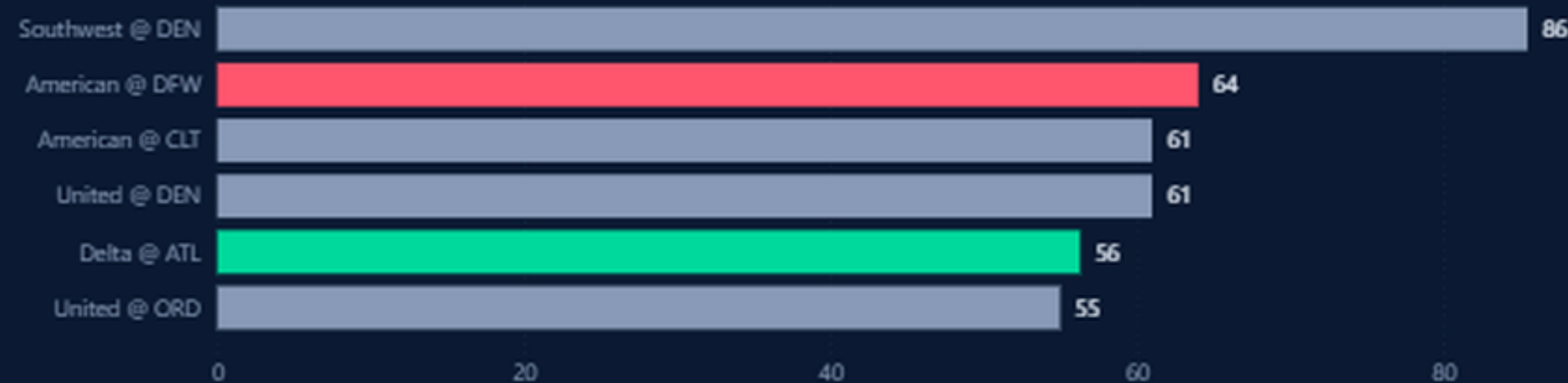
More turnaround time stops delays spreading

AA @ DFW, plane arrived 15+ min late: % that left late too



American passes on more delays than rivals

% of late arrivals passed on to the next departure, by hub



Last year's lateness predicts this year's

2025 inbound late %, by 2024 risk tier



Break-even cost by strategy

\$ per added ground minute needed to pay off



Recommendation

- 1 Skip blanket buffers: they would lose ~\$20M a year.
- 2 Pilot buffers on ~6,100 turns whose inbound flight was late 40%+ of the prior year.
- 3 Fix chronically late inbound flights at the source.
- 4 Measure: the pilot pays off if knock-on savings reach 1.37x.